RISHI PRANESH. R

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Professional Summary

- 2.6 years of professional experience as **Member Technical Staff** at **HCL Technologies Limited**, focused on **Embedded Software Development** in the **Automotive Domain**, principally worked on **AUTOSAR** projects. (Client-Robert Bosch Global Software Technologies)
- Experienced in **CAN Signal configuration** and **COMSTACK configuration** (**BSW Layer**).
- Validated embedded software and improved test coverage through implementation of Hardware-in-the-loop **(HIL)** and Software-in-the-loop **(SIL)** Testing methodologies.
- Well-versed with automobile industry standards like **ISO 11898, ISO 14229,** software architecture like **Classic AUTOSAR,** coding guidelines like **MISRA C** and process model like **ASPICE.**
- Collaborated with leading automobile brands like Mahindra, TATA Motors as part of the development and testing team. Acquired expertise in performing development, integration and testing of automotive modules using various tools, develop and review test cases, execute them and provide support for debugging and rectification.
- Developed and tested **Vehicle Control Driver Input functions,** employing methodologies such as **Unit Testing** to ensure software quality and reliability.
- Skilled in **Verification and Validation (V&V), Requirement Engineering** (including Tracking and Traceability using **IBM Doors**), and requirements management tools such as **IPE** and **RQONE.** Proficient in documentation using **ECU WorX.** Experienced in system level and software testing with a strong emphasis on clarity, completeness and traceability of requirements and test-artifacts.
- Possess comprehensive knowledge of communication protocols including **CAN, LIN,** and diagnostic protocols such as **UDS.**
- Proficient in **C programming language.**
- Experienced in **Software Development Life Cycle (SDLC)**, applying the **V-Model methodology** for structured development and testing.
- Efficient in managing **pre- and post-software integration activities** to support system validation and deployment.
- **Input/Output Pin (ADC, DIO, PWM) Configuration** and **Testing.**
- **AUTOSAR ASW** Component Development using **ISOLAR.** Functional overview of **DIAGSTACK.**
- Conducted CPU utilization tests and analyzed memory consumption.
- Involved in **Functional testing of ECU** and **Vehicle CAN Testing.**

Technical Skills

- **Programming Languages:** C language
- **ECU Development and AUTOSAR Tools:** ECU WorX, AUTOSAR BCT Editor, Signal Configuration tool
- **Designing and Simulation Tools:** ASCET, MATLAB (Basic proficiency)
- **CAN Testing and Diagnostics Tools:** VN1611, Vector CANalyzer, CAN Diagnostics
- **Software Testing and Debugging Tools:** INCA, Modoki, Diagra-D scan tool, Oscilloscope, Pico scope, Function generator
- **Flashing and support tools:** CFI, SDOM
- **Requirement Engineering tools:** IBM DOORS, IPE, IBM RQONE.

Work Experience

SOFTWARE ENGINEER | CLIENT: BOSCH | Location: Bangalore **AUG 2023 - JUNE 2024**
1. TATA MOTORS PASSENGER VEHICLES – NEXON, ALTROZ

ROLES AND RESPONSIBILITIES:

Tracking and Traceability of requirements using DOORS, IPE or RQONE
- Developed, Integrated and Tested software for driver input control functions, improving system performance and software efficiency across TATA Nexon and Altroz platforms.
- Designed and configured CAN frames and signals, conducting functional testing using Vector CANalyzer.

- Integrated software modules or packages from cross-functional teams into a software stack for final stages of delivery.
- Analyzed and debugged issues reported from OEM, eliminating it in further releases.
- Resolved build-time compilation errors and analyzed log files to identify and fix issues during testing as part of the software build toolchain.
- Configured **DID**s using **Signal Configuration Tool** and contributed to automated build processes and test scenario development.
- Performed configuration and testing of I/O pins including **ADC, DIO, and PWM**, ensuring accurate signal interpretation and hardware interfacing.

SOFTWARE ENGINEER | CLIENT: BOSCH | Location: Bangalore FEB 2022 – AUG 2023
2. TATA MOTORS COMMERCIAL VEHICLES – ACE, INTRA BI-FUEL

ROLES AND RESPONSIBILITIES:

Integrated and Tested software for driver input control functions across TATA ACE and INTRA platforms.

- Executed various testing activities including:
- **Unit Testing**, **Configuration Testing**, **Functionality Testing**
- **Pre- and Post-Integration Testing**
- **Resource Measurement Analysis** using HIL setups
- **Memory Consumption Analysis** through drive-cycle tests on both HIL and in-vehicle environments.
- Supported troubleshooting and debugging issues during build and handled software flashing issues.
- Exposed to source-level debugging and profiling.
- Involved in Functional testing of ECU and Vehicle CAN testing.
- Reproduced, Investigated, and follow up on defects found during validation. Drove sightings from open to closure by aligning with different teams.

Education

BACHELOR OF ENGINEERING IN ELECTRONICS AND COMMUNICATION | 2017-2021 | SNS COLLEGE OF ENGINEERING, COIMBATORE, TAMILNADU | 7.26 CGPA

Language

Tamil (Native), English (Professional), Kannada (Elementary), Telugu (Elementary)

Certifications

Generative AI and Prompt Engineering for Absolute Beginners, Udemy – September 2025

* Developed Core competencies in Generative AI concepts, mastering prompt engineering, Custom GPT's, Gen AI for Business productivity, Hallucinations, AI for Social Media Marketing, and Automating workflows.

Become a Machine Learning Expert: Introduction to MATLAB, Simplilearn – October 2025

* Introduction to MATLAB - Variables, Datatypes, Operators, Scripts and Functions overview.

Awards and Recognition:

Awarded ERS (Engineering and R&D Services) Process Champion by HCL for strong commitment to work and exceptional support in critical project deliveries.